

Ecommerce Consumer Behavior Analysis Data

Executive Summary:

This analysis provides comprehensive insights into customer purchase behavior, demographics, loyalty, and decision-making patterns to support data-driven business strategies.

Scope

Purchase Behavior:

- 1-Total Purchase Amount? 2-Total Orders ? 3-average customer purchase frequency?**
- 4-purchase order accuracy?**
- 5-purchase frequency , Purchase Amount by purchase channel , purchase intent , Shipping Preference , Device Used for Shopping , Payment Method and Social Media Influence.**
- 6-Purchase Frequency, Purchase Amount by Discount use, Discount Sensitivity .**
- 7-purchase frequency by months?**

Customer Demographics: target audience

- 1-percent of customers by gender ,age .**
- 2- Purchase amount by Education Level income level.**
- 3- top 10 customer and locations by purchase amount.**

Customer Loyalty:

- 1-Average rating of Customer Satisfaction, Brand Loyalty .**

Making decision:

- 1- Average of Time Spent on Product Research(hours), Time to Decision ,CLV.**

Data Collection & Methodology:

- Data source**: Ecommerce_Consumer_Behavior_Analysis_Data (csv file).
- Tools Used** : Power BI(power query , dax).
- Pre processing steps** :add new measures by dax, replace values in gender column,Split and change data type of column(purchase amount) by power query.

Data Analysis:

Purchase Behavior:

1-Total Purchase Amount? 2-Total Orders ? 3-average customer purchase frequency?
4-purchase order accuracy?

Average Order Value	purchase accuracy	Estimated Customers	purchase frequency
\$272.54	86%	1.00K	7

Insight:

- 6,945** order with a AVO of **273\$** in one year show **strong demand** .
- 86%** accuracy mean our strategies are **reliable** .
- we have **around 1000** customer actively contributing to our revenue.
- customers purchase **7 times** on average

Recommendations:

- 1- Maintain and Enhance this value by promoting premium products.
 - 2- Recommend products based on customer interest to drive upselling.
 - 3- Aim to increase accuracy to 95%:
 - Try to get customer's feedback(know disadvantage of our system).
 - Improve our order tracking system .
 - 4- Keep existing customers engaged and motivate them by personal offers, build a customer community.
 - 5- Recommend more then 7 products based on each customer interests.
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5-Purchase Frequency, Purchase Amount by Discount use, Discount Sensitivity ?



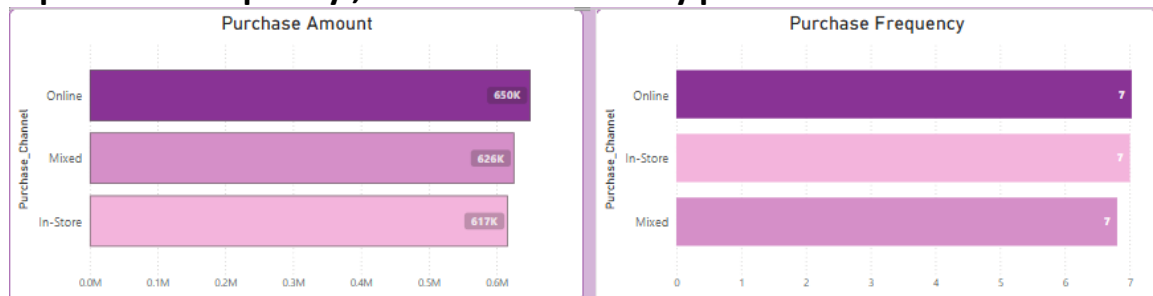
Insight:

- purchase amount **increase by 4%** from not sensitive to very sensitive so discount affect purchase moderately .
- discount **don't increase** how often people buy.

Recommendations:

- 1- Target only discount sensitive customers .

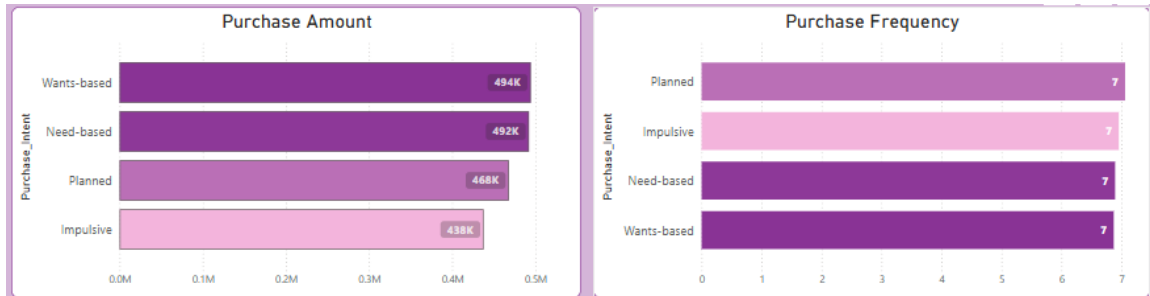
6-purchase frequency , Purchase Amount by purchase channel ?



Insight:

- **online purchase channel** has high contributions in purchase amount (**34%** of total purchase amount) then mixed purchase channel in second level (**33%** of total purchase amount) then in-store purchase channel (**32%** of total purchase amount).
- very small change in frequency , has no effect .

7-purchase frequency , Purchase Amount by purchase intent ?



Insight:

- purchase amount in **increase** if user wants or need some product (users spend more depend on their demand).
- purchase Frequency in **increase** if user are plane or impulsive .

Recommendations:

- 1- Focus on build trust and loyalty by (provide good quality , easy of communication)
- 2- Use storytelling as way to promote products to attract customers

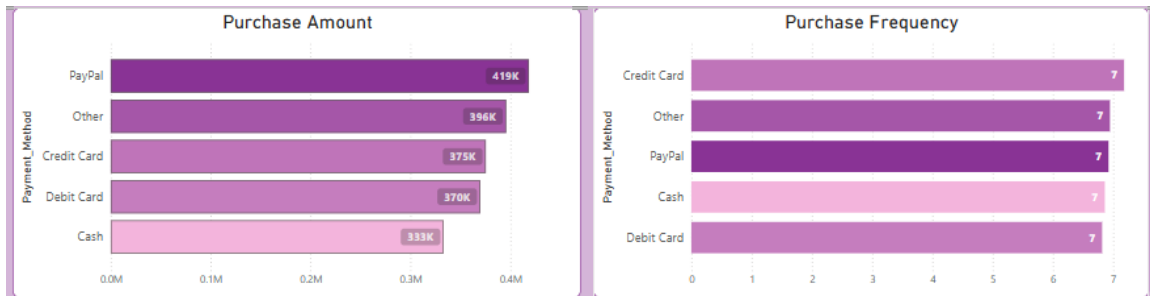
8-purchase frequency , Purchase Amount by Shipping Preference ?



Insight:

- no preference shipping increase purchase amount .
- Express purchase is most common .

9-purchase frequency , Purchase Amount by Payment method?



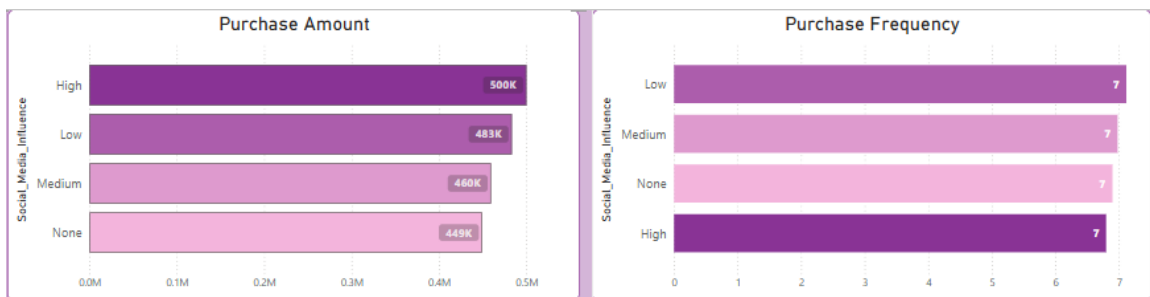
Insight:

- customers who pay with **pay pal** have the **highest** purchase Amount (all other methods around 390K to 333K).
- customers who pay with **credit card** have the **highest** purchase Frequency (all other methods around 6.6 to 6.8).

Recommendations:

- 1- Encourage credit card and pay pal usage where possible by offer exclusive benefits, highlight credit card options

10-purchase frequency , Purchase Amount by Social Media Influence?



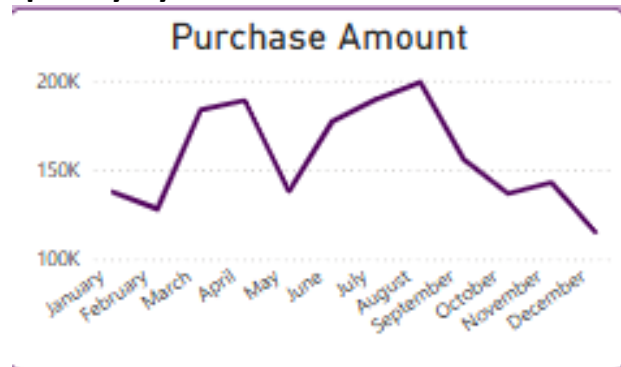
Insight:

- purchase amount in **increase** by **high influence** of Social Media (high influence user spend more per orders).
- purchase Frequency in **increase** by **low influence** of Social Media (This mean that low influence users are more loyal or habitual buyers).

Recommendations:

- 1- for **high influence** I recommend focus on influencer campaigns ,improve ticket for product highlights.
- 2- for **low influence** I recommend give them priority customer service.

9-purchase frequency by months?



Insight:

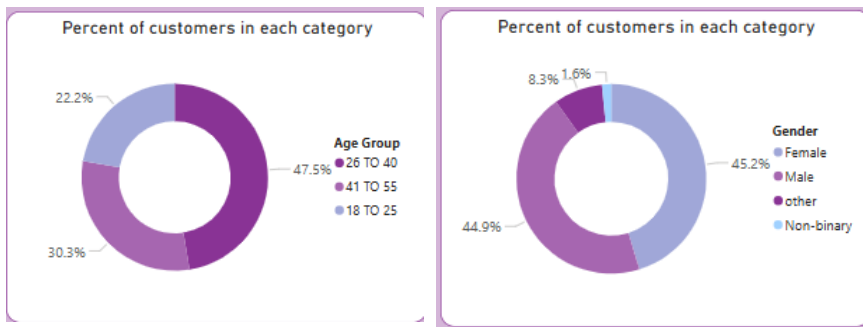
- purchase amount increase more in quarter 3 (July ,august ,September) and (march ,april ,may)(Mean that purchase frequency are high).
- purchase amount decrease more in quarter 4(oct , nov , dec) and (Jan ,feb ,jun)(Mean that purchase frequency are low).

Recommendations:

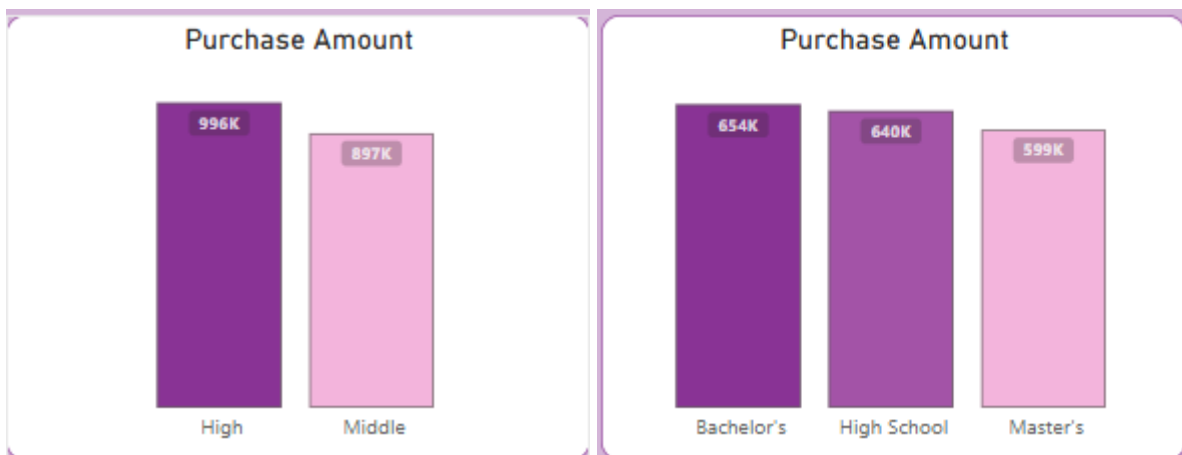
- 1- for high purchase amount months optimize Inventory (Ensure stock availability for their frequently purchased items).
- 2- Offer promotions or product launches to promote purchase in other months.

Customer Demographics: target audience:

1-percent of customers by gender ,age ?



2-purchase amount by Education Level, income level?



Insight:

- Our service are **for all ages and all gender** .
- customer with **high income** level spend **more than middle** income level.
- **bachelor's education** level spend **more than high school** level then masters.

Recommendations:

- 1- Target high income level, bachelor's education level customer .

3-top 10 customer and locations by purchase amount?

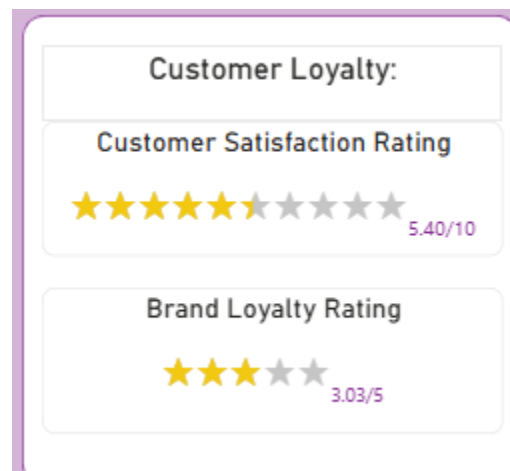
Customer_ID	Purchase Amount	Rank	Location	Purchase Amount	Rank
69-394-1424	5,880.24	1	Muff	5,880.24	1
82-671-7236	5,872.76	2	Ābdānān	5,872.76	2
90-144-9193	5,836.80	3	Liren	5,836.80	3
42-518-2602	5,573.77	4	Navoiy	5,573.77	4
25-286-8786	5,541.72	5	Berlin	5,541.72	5
83-427-3218	5,495.68	6			
Total	55,689.18	1	Total	28,705.29	1

Recommendations:

- 1- I recommend Personalizer offers , gifts ,exclusive support for top 5 customers .
- 2- Plan good strategy for inventory to reduce delivery time.

Customer Loyalty:

1-Average rating of Customer Satisfaction, Brand Loyalty ?



Insight:

- While 3.03/5 loyalty is decent, it's not very strong — and combined with only 5.40/10 satisfaction, **it could mean loyalty is weak.**

Recommendations:

- 1- se surveys, reviews, or direct interviews to find pain points.
- 2- Enhance Product/Service Quality.

Making decision:

- 1- Average of Time Spent on Product Research(hours), Time to Decision ?
- 2- Customer lifetime value?

No. of customers	Customer lifetime value	Time Spent on Product Research	Time to Decision
1000 	1.89K	1 Hours	8 Days

Insight:

- Quick Research, Slow Decision.
- Despite limited research time, the CLV is high (1890).

Recommendations:

- 1- Send follow-up emails or reminders within 2–4 days of product interest.
- 2- Offer time-sensitive incentives (e.g. “Buy within 48 hours and get 10% off”) to nudge quicker decisions.

“implementing these data-driven strategies, we aim to optimize customer engagement, increase conversion rates, and maximize purchase . We look forward to further discussions on refining these strategies and executing the next phase of our growth plan”

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